SPARK CODE6

SMART INDIA HACKATHON

SMART INDIA HACKATHON 2026

- **Problem Statement ID – 26237**

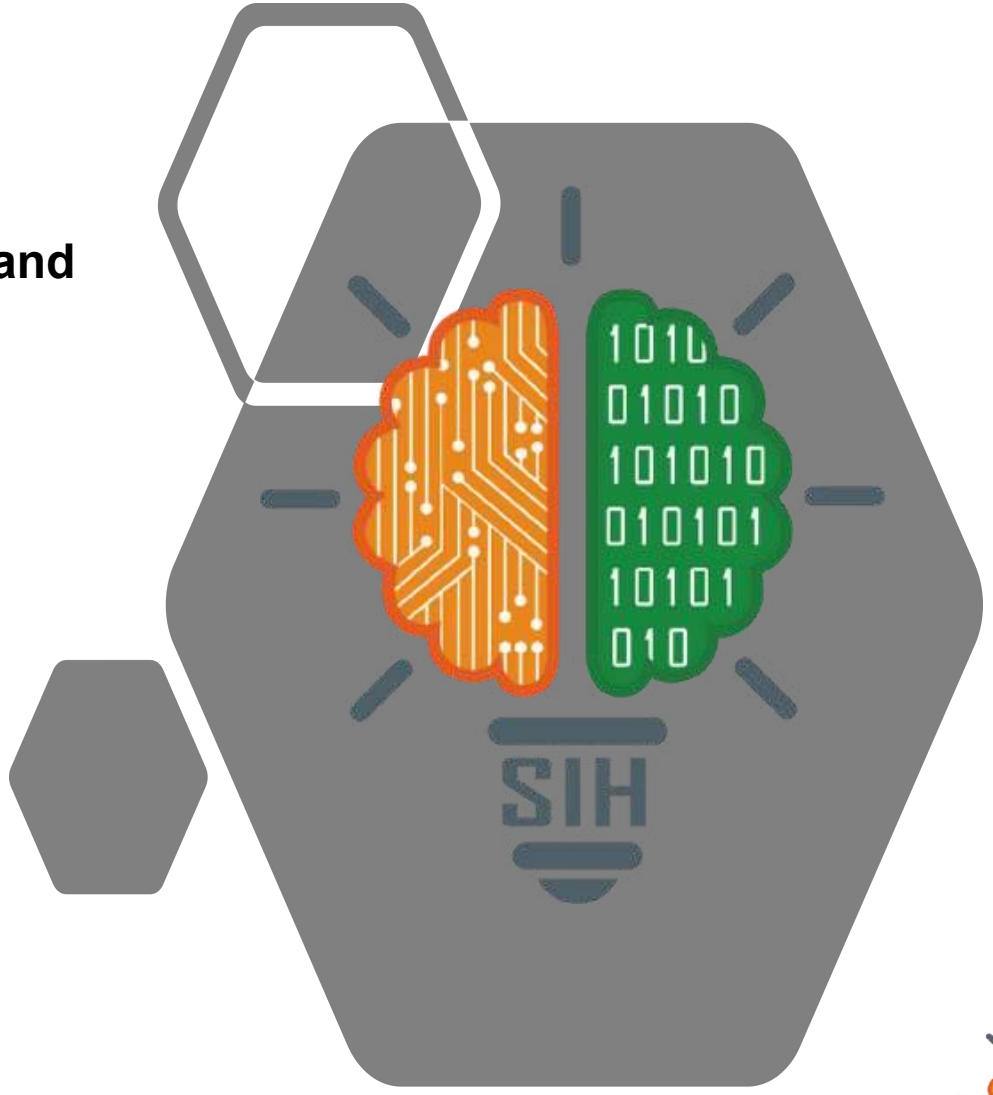
- **Problem Statement Title - Cryptographic Attribution and Immutable Decryption Provenance for Multi-Recipient Encrypted Document Distribution**

- **Theme- Blockchain & cybersecurity**

- **PS Category- Software**

- **Team Name – SPARK CODE6**

- **Team ID - 164931**



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CHAKRAVYUHA PQC-

Immutable Post-Quantum Decryption Provenance with Invisible Forensic Watermarking Console having a Sovereign Post-Quantum Broadcast Encryption & Forensic Leak Attribution System







PROBLEM	REMEDY
1 Multi-Recipient Encrypted Documents Are Identical Upon Decryption, Making Desktop Screenshots And Photos Completely Untraceable	1 Dynamic Dual-Layer Invisible Forensic Watermarking Embedded In Both PDF Text Streams And Screen Pixels At Decryption
2 Centralized Database Decryption Logs Can Be Silently Modified Or Deleted By Privileged Administrators To Frame Innocent Colleagues	2 Air-Gapped Immutable DLT Blockchain Secured By Multi-Command Proof-of-Authority Consensus And SHA3-256 Merkle Inclusion Proofs
3 Classical Encryption Fails Against Quantum Computers, While Standard Viewers Allow Leakers To Deny Ever Decrypting The Document	3 Native NIST FIPS 203/204 Post-Quantum Cryptography Enforcing Mandatory Digital Signatures Before Plaintext Is Unlocked
4 Defense Command Centers And Warships Are Strictly Air-Gapped And Cannot Depend On Commercial Cloud KMS Or Public Blockchains	4 Fully Sovereign Air-Gapped Architecture Executing 100% Offline With Local Keystores And Zero External Cloud Dependencies

















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TECHNICAL APPROACH

TECHNOLOGICAL STACK OF "CHAKRAVYUHA PQC "

User & Operational Layer

- Hardware TPM 2.0
- Air-Gapped PKI Keystore
- Military RBAC Clearance

Frontend Layer

- PyWebView Desktop App
- Tailwind CSS UI
- HTML5 Canvas API

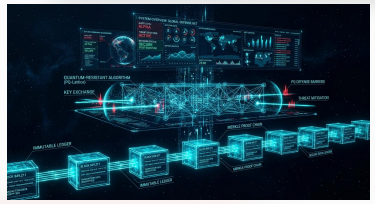
Backend & Cryptographic Engine Layer

- Python 3.11 Core
- NIST PQC Algorithms
- AES-256-GCM Encryption

Data & Immutable Ledger Layer (DLT)

- SHA3-256 Merkle Trees
- Consortium PoA Blockchain
- Encrypted JSON-RPC Ledger





WHY WE ARE UNIQUE ???

Anti-Collusion Traitor Tracing That Exposes Both Officers If They Merge Documents To Erase Watermarks

Moiré-Resistant Perspective De-Warping That Extracts Invisible Watermarks Even From Tilted Smartphone Screen Photos

Micro-Syntactic Semantic Honeytokens That Identify Leakers Instantly From Quoted Text Snippets On Social Media



Zero-Knowledge Attribution Proofs That Legally Prove Treason Without Declassifying Or Revealing Top-Secret Document Contents

Hardware TPM 2.0-Sealed Ephemeral RAM Decryption That Auto-Destructs If Copied To Unauthorized USB Drives

Automated Court-Martial Evidentiary Dossier Generator Fully Compliant With Section 65B Of The Indian Evidence Act

GIT HUB REPOSITORY -

https://github.com/mrad11/CHAKRAVYUHA-PQC

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FEASIBILITY & VIABILITY

- NIST FIPS 203 & 204 Standard
- Section 65B Indian Evidence Act

- CERT-In / ISO/IEC 27001 Alignment
- 100% indigenous cryptographic architecture

FEASIBILITY



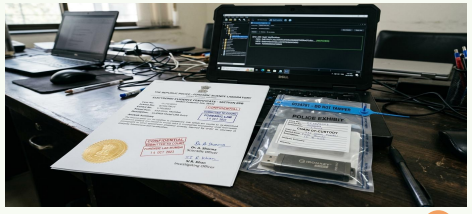
Zero-Hardware-Overhead Deployment on Existing Military Infrastructure with Decades of Post-Quantum Sovereign Resilience

Battlefield-Ready Air-Gapped Feasibility Delivering Mathematically Unbreakable National Defense Viability for 30+ Years

Flawless Offline Operational Feasibility Coupled with Legally Enforceable Deterrence Across All Tri-Service Commands



VIABILITY



Instant Field Deployment on Existing Air-Gapped Laptops Without Costly Hardware or Infrastructure Overhauls

Long-Term Sovereign Independence Eliminating Foreign Software Licenses and Multi-Crore Breach Disasters

Thirty-Year Quantum Immunity Aligned with Global FIPS Standards and Section 65B Judicial Admissibility

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IMPACTS & BENEFITS



IMPACTS

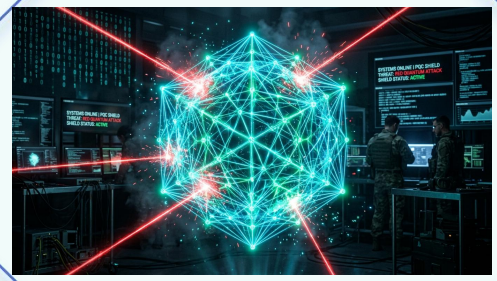
BENEFITS



Immunizing India's 30-Year Military Strategy Against Quantum Supercomputing Decryption

Instant Cross-Service Classified Document Sharing Across Bunkers, Submarines, and High-Altitude FOBs

Dismantling the Anonymity Shield to Eliminate Rogue Insider Leaks Before They Occur



Transforming Circumstantial Leaks into Legally Incontestable Evidence Within Seconds

Eliminating Foreign Proprietary Black-Boxes, Supply-Chain Kill-Switches, and External Leverage

Commercial-Grade Speed on Standard Military Laptops Without Buying Costly Hardware Accelerators

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RESEARCH & REFERENCES

https://www.cert-in.org.in/



https://indiacode.gov.in/home



https://csrc.nist.gov/projects/post-quantum-cryptography



https://datatracker.ietf.org/doc/html/rfc5116



Feature Dimension	Conventional Tools (Adobe DRM, PGP, GPG)	CHAKRAVYUHA PQC (Live Today)	Next-Gen Innovation (Adding Next)
Quantum Threat	✗ Easily cracked by future quantum computers (RSA/ECC)	✓ NIST FIPS 203 & 204 lattice math unbreakable by quantum PCs	🚀Crypto-agility with 50-year quantum key longevity
Air-Gapped Ops	✗ Fails offline; strictly requires internet license servers	✓ 100% offline readiness in bunkers with zero network dependency	🚀Tactical radio sync over military VHF/UHF frequencies
Leak Attribution	✗ Zero proof; cannot tell which officer leaked a shared file	✓ Signed blockchain receipt (DPR) minted per officer view	🚀Anti-collusion math catching traitors even if copies are merged
Phone Screen Cam	✗ Defenseless against an officer photographing the screen	✓ Invisible spatial watermark surviving mobile camera photos	🚀Neural moiré filter surviving extreme WhatsApp compression
Insider Espionage	✗ Passive trust relying entirely on NDAs and honor codes	✓ Officer rank, ID, and timestamp forensically bound to pixels	🚀Automated canary trap catching rogue moles in seconds
Legal Evidence	✗ Circumstantial server logs easily dismissed in court	✓ Tamper-proof SHA3-256 Merkle proofs with MoD validator signatures	🚀TPM 2.0 silicon lock restricting access strictly to military hardware